Fall 2020

Problem 1 (Fa20). Find the indefinite integral $\int e^{2x} \sin x \, \mathrm{d}x$.

Problem 2 (Fa20). Let S be a set and let $\{f_n\}$ and $\{g_n\}$ be sequences of functions $S \to \mathbb{R}$.

1. Show that if $\{f_n\}$ and $\{g_n\}$ converge uniformly to bounded functions f and g on S, respectively, then $\{f_n g_n\}$ converges uniformly to fg on S.

2. Show, by giving a counterexample, that the statement is false if f is not required to be bounded.

Problem 3 (Fa20). Let X be a metric space, and let $\{T_1, T_2, \ldots\}$ be an infinite sequence of nonempty closed subsets of X. Assume that T_1 is compact and that $T_n \supseteq T_{n+1}$ for all $n \geqslant 1$. Show that

$$\bigcap_{n=1}^{\infty} T_n \neq \emptyset$$

Problem 4 (Fa20). Find the set of complex numbers z for which the infinite product

$$\prod_{k=1}^{\infty} \left(1 - z^k\right)$$

converges.

Problem 5 (Sp15, Fa20). Write two different Laurent series in powers of the complex variable z for the function

$$f(z) = \frac{1}{z\left(1 + z^2\right)}$$

Give the domain of each of these series.

Problem 6 (Sp83, Fa21, Sp20, Fa20). Let A be an invertible real $n \times n$ matrix. Show that there is a decomposition $A = QR$ in which Q is an $n \times n$ real orthogonal matrix and R is upper triangular with positive diagonal entries. Is this decomposition unique?

Problem 7 (Sp20, Fa20, Fa25). Let V be a vector space of dimension n over a finite field with q elements. Prove that the number of subspaces $W \subseteq V$ of dimension k is equal to

$$\frac{\prod_{j=1}^{n}\left(q^j-1\right)}{\left(\prod_{j=1}^{k}\left(q^j-1\right)\right)\left(\prod_{j=1}^{n-k}\left(q^j-1\right)\right)}$$

Problem 8 (Fa20). Let $\mathbb{Z}_{179}$ be the finite field with 179 elements.

1. Prove that the residue class of $10 \pmod{179}$ is not the square of any element of $\mathbb{Z}_{179}$.

2. Prove that this residue class generates the multiplicative group of nonzero elements of $\mathbb{Z}_{179}$.

Problem 9 (Fa20). Prove that if a and b are odd integers, then the polynomial $f(x) = x^3 + ax + b$ has no rational zeros.

Problem 10 (Fa20). Prove that it is not possible to find 4 polynomials $a(x)$, $b(x)$, $c(x)$, $d(x)$ with real coefficients such that

$$a(x) < b(x) < c(x) < d(x) \qquad \text{for} \quad 0 < x < 1$$

and

$$b(x) < d(x) < a(x) < c(x) \qquad \text{for} \quad -1 < x < 0$$

Problem 11 (Fa20). Either prove or give a counterexample to each of the following statements:

1. If the series of real numbers $a_1 + a_2 + \cdots$ and $b_1 + b_2 + \cdots$ are both convergent then so is $(a_1 + b_1) + (a_2 + b_2) + \cdots$

2. If the series of real numbers $a_1 + a_2 + \cdots$ and $b_1 + b_2 + \cdots$ are both convergent then so is $a_1b_1 + a_2b_2 + \cdots$

Problem 12 (Fa20). Prove that every closed subset C of the real line is the closure of a finite or countable set.

Problem 13 (Fa12, Fa17, Fa20).

1. Find the poles and residues of $\dfrac{1}{z^3 \cos z}$.

2. Show that the integral of the function above over a square contour centered at the origin with side $2\pi N$ tends to zero as the integer N tends to infinity.

3. Find the sum $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots$

Problem 14 (Fa98, Fa20). Let f be analytic in the closed unit disc, with values $f(-\log 2) = 0$ and $|f(z)| \leqslant |e^z|$ for all z with $|z| = 1$. How large can $|f(\log 2)|$ be?

Problem 15 (Fa20). Let

$$H_{ij} = \int_0^1 t^i t^j \, \mathrm{d}t \qquad 0 \leqslant i, j \leqslant n$$

be the elements of the $(n+1) \times (n+1)$ Hilbert matrix H. Let

$$P_i(t) = \sum_{j=0}^{i} p_{ij} t^j \qquad 0 \leqslant i, j \leqslant n$$

define the coefficients p_{ij} of the orthonormal Legendre polynomials so that

$$\int_0^1 P_i(t) P_j(t) \, \mathrm{d}t = \delta_{ij} \qquad 0 \leqslant i, j \leqslant n$$

Show that $H^{-1} = P^{\top} P$, where P is the matrix with entries p_{ij}.

Problem 16 (Fa20). Let A be a linear transformation on a vector space W over a field $\mathbb{F}$, such that $A^5 = I$.

1. Show that if $\mathbb{F}$ does not have characteristic 5 then W can be written as a direct sum of subspaces U and V where U consists of the vectors u with $Au = u$, and $AV = V$.

2. Give an example to show that this property can fail if $\mathbb{F}$ has characteristic 5.

Problem 17 (Fa20). Find all conjugacy classes of S_3.

Problem 18 (Fa20). For which integers n is $x^4 + n$ a reducible polynomial in $\mathbb{Z}[x]$?